

Hospital Pathway Group Acceptance Test Plan and Results

Submission: AISD (UFCEP6-0-3) - Acceptance Test Plan and Results

Team: Ruby (A), Estrella (B), Ender (C), Ryan (D)

Formal model: Hospital_All_Processes_Simple_C8

Repository baseline reviewed: fc6293e, 29 September 2026

Test environment: local Camunda 8 c8run, Tasklist/Operate, Java 21, Maven, Spring Boot workers, file-based H2 domain database

Test data: fictional case identifiers only

Purpose and verdict rules

This is the group's consolidated plan and result record for the executable hospital pathway. Each scenario states its preconditions, data, actions, measurable expected outcome, pass/fail rule, actual observation and evidence. It combines existing group plans with PB-14 and PB-18 execution records, current Java tests and one live final-model start-to-end run. The other final-model branches remain separately assessed.

PASS means all stated checks in the recorded run matched the expected outcome. **FAIL** means at least one check did not match. **NOT EVIDENCED** means the repository has no execution record for that scenario at the final baseline; it is not a pass. **PASS, LIMITED SCOPE** means the named checks passed, but a wider business rule still needs a separate check. A result from deployed process version 6 is labelled historical and must be repeated if used to claim the final version.

The acceptance route uses the formal English package Coursework/hospital-pathway/. Start c8run, then launch Java from that module directory with `mvn spring-boot:run. ./data/hospital-domain` is the application database, not the Camunda engine database. A process started without Java may wait at a service task. The role in the form is a recorded value; current Tasklist tasks are assigned to `demo` and do not enforce staff authority.

Evidence register

E1

Submitted source: Ryan/2026-09-26/PB-14_acceptance-run-log_2026-09-29.json and matching PB-14 plan

Baseline and evidential limit: Real local Camunda REST /v2, deployed definition version 6. G01/G02 routing and variables only; G01 stops at clinical care.

E2

Submitted source: Ryan/2026-09-26/PB-18_acceptance-run-log_2026-09-29.json and matching PB-18 plan

Baseline and evidential limit: Real local Camunda REST /v2, deployed definition version 6. G03 fail, G04/G05/G06 pass on their listed checks.

E3

Submitted source: Coursework/hospital-pathway/evidence/Java_Test_Results_2026-09-29.txt; A's RefundWorkersTest.java; D's CareChangeWorkersTest.java

Baseline and evidential limit: Clean Maven run at repository commit fc6293e; A five and D seven Java tests passed, zero failures/errors. Uses in-memory H2 and mocked incoming Camunda jobs. This result does not test B or C worker behaviour.

E4

Submitted source: Coursework/hospital-pathway/evidence/BPMN_Structural_Test_2026-09-29.txt

Baseline and evidential limit: Four Python structural checks at fc6293e; checks model references and worker types, not a live deployment.

E5

Submitted source: Est/est: 全院流程路径记录.md; Est/est: 治疗预约与付款五条路径记录.md

Baseline and evidential limit: B's historical engine observations: six whole-hospital version-1 routes and five earlier booking/payment-model cases. They are not runs of the current formal BPMN; the earlier model also records an invalid-input incident and a repeated-payment case waiting at a message.

E6

Submitted source: Est/est: 领域数据库设计.md; formal BookingWorkers.java and domain/ source

Baseline and evidential limit: B's H2 schema and three booking workers are present. Design/source evidence, not a B-specific Java test or final-version booking result.

E7

Submitted source: Ender/2026-09-28/C_外部工作器与数据库说明.md; formal LetterWorkers.java and the two corresponding BPMN service tasks

Baseline and evidential limit: C's letter-dispatch and referrer-notification implementation and mappings are present. Source evidence, not a committed C-specific behavioural test or actual message delivery.

E8

Submitted source: Ruby/hospital-pathway-zh-learn/验收测试计划-S1至S10-中英对照.md and matching PDF

Baseline and evidential limit: A/group S1-S10 acceptance scenarios and presentation plan. A plan is not a run log; the success-payment ledger expectation conflicts with the current payment worker and is corrected in G07.

E9

Submitted source: Coursework/hospital-pathway/evidence/Group_Final_E2E_20260929-135729.json; repeatable runner scripts/group_final_e2e.py

Baseline and evidential limit: Live local Camunda 8 REST /v2 run on the exact BPMN SHA-256 recorded in the JSON; deployed definition version 1, instance 2251799813685458, nine user-task completions, final COMPLETED, no active incident, ten true assertions and six captured worker-log lines. This proves the selected self-pay route. REST task completion does not by itself prove browser-side Tasklist form validation or every alternate branch.

The G07 evidence was generated locally on 29 September against source baseline fc6293e after starting c8run and the Spring Boot workers. It is included in this working tree but still needs the group's final commit

and independent review. The original S1-S10 PDF is a plan, not an execution log. Its expected successful-payment ledger row is not asserted because the current request-payment worker does not persist such a row.

Member-by-member coverage and missing proof

A / Ruby

Work represented in the formal BPMN, forms and Java: P5 clinical decision, P7 funding/refund, P8 urgent authorisation, P13 management review; request-refund, mark-payment-investigate, shared request-payment explanation

Existing evidence: E3: five passing refund/investigation Java tests. E8: S1-S10 plan. E9: live self-pay path and payment message.

Outstanding acceptance proof: G11 refund, P8 authorised/hold split, P13 report branch, Finance sign-off and second-owner review.

B / Estrella

Work represented in the formal BPMN, forms and Java: P3/P4 visit, P6 treatment booking; check-slot, reserve-appointment, flag-resource-unavailable; three-table H2 domain design

Existing evidence: E5: older engine route records. E6: current source/schema. E9: live first-visit and treatment reservation worker logs.

Outstanding acceptance proof: G09 pending-to-confirmed database assertions and second-owner review.

C / Ender

Work represented in the formal BPMN, forms and Java: P1 registration, P2 referral decision/redirect, P9 letter; dispatch-clinic-letter and notify-referrer

Existing evidence: E7: current source and implementation note. E9: live approved-letter worker logs and dispatch variables.

Outstanding acceptance proof: G10 referrer/redirect and retry checks, direct audit-row check, final-form validation and second-owner review.

D / Ryan

Work represented in the formal BPMN, forms and Java: P10 enquiries and P11/P12 modification/follow-up/cancellation; notify-care-change

Existing evidence: E3: seven passing Java tests. E4: four model structural checks. E1/E2: older group routes.

Outstanding acceptance proof: G08 final-version live notification/Finance hand-off, enquiry branches and second-owner review.

This matrix separates **implemented source**, **older engine observations**, **current automated tests** and **final-version acceptance**. None of the first three alone establishes the fourth.

Group result overview

G01

Scenario and principal owner: Accepted referral to first clinical care; C/B/A hand-off

Result: PASS, historical version 6

Evidence: E1

G02

Scenario and principal owner: Rejected referral ends without booking; C

Result: PASS, historical version 6

Evidence: E1

G03

Scenario and principal owner: Secretary-labelled acceptance is refused; group role control

Result: FAIL, historical version 6

Evidence: E2

G04

Scenario and principal owner: No visit slot returns to pending booking; B

Result: PASS, LIMITED SCOPE, historical version 6

Evidence: E2

G05

Scenario and principal owner: Failed mock payment followed by human retry; A/shared

Result: PASS, historical version 6

Evidence: E2

G06

Scenario and principal owner: Missing payment confirmation goes to investigation; A/shared

Result: PASS, historical version 6

Evidence: E2

G07

Scenario and principal owner: Full self-pay referral to completed care; A/B/C hand-offs

Result: PASS on the selected final-model route, local deployment version 1

Evidence: E9

G08

Scenario and principal owner: Clinical change notification and finance separation; D

Result: NOT EVIDENCED in live engine

Evidence: Java checks E3

G09

Scenario and principal owner: Treatment resource unavailable and reservation; B

Result: NOT EVIDENCED; code-level mismatch identified

Evidence: E6

G10

Scenario and principal owner: Approved letter dispatch and referrer notification; C

Result: NOT EVIDENCED end-to-end

Evidence: E7

G11

Scenario and principal owner: Finance-approved refund after a care change; A/D

Result: NOT EVIDENCED end-to-end

Evidence: Java checks E3

U01

Scenario and principal owner: Refund and investigation worker database behaviour; A

Result: PASS, five Java tests

Evidence: E3

U02

Scenario and principal owner: Care-change receipt, guards and retry behaviour; D

Result: PASS, seven Java tests

Evidence: E3

U03

Scenario and principal owner: Booking worker persistence and pending-to-ready transition; B

Result: NOT EVIDENCED; source-level mismatch identified

Evidence: E6

U04

Scenario and principal owner: Letter/referrer worker receipts and retry; C

Result: NOT EVIDENCED behaviourally

Evidence: E7

H01

Scenario and principal owner: Earlier booking/payment and whole-hospital routes; B

Result: HISTORICAL OBSERVATIONS only

Evidence: E5

S01

Scenario and principal owner: BPMN references and Java job-type alignment; group

Result: PASS, four structural checks

Evidence: E4

Recorded Camunda acceptance scenarios

G01 Accepted referral reaches clinical care

Business rules: BR-01, BR-02, BR-03. **Preconditions:** Camunda definition version 6 deployed; check-slot worker running; a unique case ID. **Data:** case PB14-20260929-004206-MAIN, request referral, consultant decision **accept** with non-empty reason and clinician ID, visit outcome **attended**. **Actions:** start process; complete Register; complete Review referral; complete Book visit. **Expected/pass rule:** Book visit appears only after acceptance; **referralDecision=accept** and reason are retained; after slot checking, Clinical care appears; no rejection end and no incident. Fail if any check is absent. **Actual:** all five PB-14 checks passed; instance 2251799813925318 remained open at Clinical care. **Verdict:** PASS for this slice on version 6; not an end-to-end pass. **Evidence:** E1.

G02 Rejected referral ends without a visit

Business rules: BR-02 and BR-03. **Preconditions:** Camunda definition version 6 deployed; unique case ID; Java worker not required. **Data:** case PB14-20260929-004206-REJECT, request **referral**, decision **reject**, reason recorded. **Actions:** complete Register and Review referral; inspect Tasklist and Operate. **Expected/pass rule:** instance is COMPLETED at rejection, decision and reason persist, no Book visit is created and no active incident exists. Fail if a booking appears or instance does not end. **Actual:** all four checks passed; instance 2251799813925262 completed. **Verdict:** PASS on version 6. **Evidence:** E1.

G03 Unauthorised secretary decision

Business rule: BR-37; PB-22. **Preconditions:** Camunda definition version 6; classroom demo account. **Data:** case PB18-20260929-011021-UNAUTH; **staffRole=Medical Secretary** / not authorised to accept; referral decision **accept**. **Actions:** register a referral, then submit Review referral using that staff role; inspect the next task. **Expected/pass rule:** the engine must refuse the unauthorised acceptance and must not create Book visit. Fail if Book visit appears. **Actual:** Book visit was created; instance 2251799813927118. **Verdict:** FAIL. The **staffRole** string is not an access-control mechanism; all tasks use **demo**. **Handling:** implement and test role restrictions under PB-22, or explicitly declare the classroom limitation in the presentation. Do not relabel this test as passed. **Evidence:** E2.

G04 No suitable first-visit slot

Business rules: BR-07 and BR-11. **Preconditions:** accepted referral at Book visit; Java check-slot worker active. **Data:** case PB18-20260929-011021-PENDING; visit outcome **pending**. **Actions:** complete Book visit as pending; wait for worker and inspect open tasks and variables. **Expected/pass rule for this run:** Book visit reappears, Clinical care is absent, **visitOutcome=pending**, no active incident. **Actual:** all four routing checks passed; instance 2251799813927181. **Verdict:** PASS, LIMITED SCOPE, version 6. The test did not query **booking_slot**, so it does not prove that no slot was marked **CONFIRMED**. Current check-slot source

requests CONFIRMED without checking `visitOutcome`; a pending visit therefore needs a direct database assertion and code review before the full business rule can pass. **Evidence:** E2 and E6.

G05 Failed payment then deliberate retry

Business rules: BR-14 and BR-17. **Preconditions:** patient route has reached Funding; Java workers active; use a unique case. **Data:** case PB18-20260929-011021-PAYFAIL; mock amount 10.00 first, then 10.01.

Actions: select patient payment; observe failed result and Resolve funding issue; record the human action; reassess funding with 10.01. **Expected/pass rule:** first result reaches the funding issue without treatment booking; after an explicit human retry, `payment_status=successful` and Book treatment appears. Fail if booking occurs after failure or cannot be reached after the deliberate retry. **Actual:** the investigation route was reached after failure, and Book treatment appeared after retry; instance 2251799813927283. **Verdict:** PASS for routing on version 6. This is a simulated provider; the test does not prove external no-duplicate-charge behaviour. **Evidence:** E2.

G06 Payment may have been taken but confirmation is missing

Business rule: BR-18. **Preconditions:** Funding task available; mark-payment-investigate worker active. **Data:** case PB18-20260929-011021-NOCONF; choose investigate. **Actions:** complete Funding, wait for worker and inspect variables and next task. **Expected/pass rule:** `payment_status=investigate`, `payment_investigate=true`, Funding issue waits for a human, and Book treatment does not open automatically. Fail on any automatic booking or successful-payment assumption. **Actual:** all four checks passed; instance 2251799813927534. **Verdict:** PASS on version 6. **Evidence:** E2.

Final-version group scenarios

G07 now has a recorded live result. G08-G11 retain measurable pass/fail rules but are **NOT EVIDENCED** as complete final-model scenarios; a G07 pass does not transfer to their different branches.

G07 Complete self-pay pathway to an end event

Owners and rules: A/B/C hand-offs; BR-02, BR-03, BR-08, BR-12, BR-14, BR-16, BR-30. **Preconditions:** the final BPMN and eleven forms deployed together; one Spring Boot application running from Coursework/hospital-pathway/; unique fictional case ID. **Data:** GROUP-E2E-20260929-135729; referral accept; visit attended; clinical plan treatment; letter sent; funding `patient_payment_required`; mock amount 10.01; treatment resources ready; final clinical plan discharge. **Actions:** the repeatable runner started an instance and completed nine Camunda user tasks via REST /v2 with variables matching the bound forms; it then queried instance state, variables and incidents and captured worker-log excerpts. **Expected/pass rule:** the payment worker publishes correlated `payment-result`; `payment_status=successful` and a transaction reference are present; B reservation and booking-confirmation workers run; C letter dispatch records a mock receipt; the instance reaches the care-ended route with state COMPLETED and no active incident. Fail if it stalls, misses these signals or ends elsewhere. **Actual:** deployed definition Hospital_All_Processes_Simple_C8, version 1, instance 2251799813685458 reached COMPLETED; nine tasks completed; all ten recorded assertions were true; zero active incidents. Worker logs show check-slot, dispatch-clinic-letter (twice), request-payment publishing `payment-result`, reserve-appointment and send-booking-confirmation; variables include `booking_slot_status=CONFIRMED`, `clinicLetterDispatchStatus=dispatched`, `payment_status=successful` and `confirmation_sent=true`. **Verdict:** PASS for this one final-model self-pay route (E9). This was API-driven task completion, not a browser Tasklist walkthrough; direct H2 row counts,

form validation in the UI, role enforcement and all alternate routes need separate tests. No successful-payment `payment_ledger` row is claimed.

G08 Care change notification and Finance hand-off

Owner and rules: D with A/Finance; BR-22, BR-24 and BR-38. **Preconditions:** final model and Java running. **Data:** case GROUP-CHANGE-001; Register request type `change`; clinician ID recorded; Clinical change action `discharge_paid`. Use a second case with action `invalid`. **Actions:** submit the authorised change; inspect `notify-care-change` result, audit row and next Finance task. Then submit the invalid case and inspect the rework route. **Expected/pass rule:** paid authorised change returns `SENT_FINANCE_REVIEW_REQUIRED`, one keyed audit receipt and a Finance review; the worker itself does not issue a refund. Invalid change returns `SKIPPED_UNAUTHORISED` and does not advance as authorised care. Fail on automatic refund, duplicate receipt on same job retry or invalid approval. **Actual/verdict:** NOT EVIDENCED in a committed final-version engine run. Seven D Java tests cover the worker and database only (E3).

G09 Treatment resource unavailable and subsequent reservation

Owner and rules: B; BR-10 and BR-11. **Preconditions:** funding/authorisation complete; booking workers running; clean case-specific `booking_slot` state. **Data:** case GROUP-SLOT-001; first Book treatment action `pending`, later `ready`, with the same time window. **Actions:** submit `pending`; inspect open task, `booking_slot` status and retry count; then submit `ready` and inspect reservation/confirmation. **Expected/pass rule:** `pending` resources keep the booking unresolved with a `PENDING` record, no premature confirmation, and a retry count; `ready` resources update the one case/window occupancy record to `CONFIRMED` and allow confirmation. Fail if a `pending` resource is marked `confirmed`, remains `PENDING` after `ready`, or duplicate occupancy records appear. **Actual/verdict:** NOT EVIDENCED in a live run. Source inspection of `BookingWorkers.upsertBooking` shows that an existing row is only changed when the requested status is `PENDING`; a later `CONFIRMED` call reuses the row without changing its status. This is a concrete code-level mismatch to resolve and test, not a claimed observed engine failure (E6).

G10 Approved clinic letter and referrer communication

Owner and rules: C; BR-30, BR-31 and BR-32. **Preconditions:** final model and letter workers running; unique case IDs. **Data:** GROUP-LETTER-001 with approved letter action sent; GROUP-REDIRECT-001 with consultant redirect. **Actions:** progress the first case to Dispatch letter, then the second to Redirect; inspect worker logs, process variables and `audit_event`. **Expected/pass rule:** approved letter has `clinicLetterDispatchStatus=dispatched` and one mock audit receipt; redirect has `referrerNotificationSent=true` and one receipt; no clinical letter content is altered by administration. Fail if a worker job incidents, a required receipt is absent or the same retry produces duplicate receipts. **Actual/verdict:** NOT EVIDENCED in a committed final-version run. The workers record simulated dispatch only; they do not prove real delivery or the seven-day target.

G11 Finance-approved refund after a money-affected change

Owners and rules: A and D; BR-20, BR-21 and BR-24. **Preconditions:** final model and workers running; a change has reached the Finance review; clean case-specific ledger/audit state. **Data:** case GROUP-REFUND-001; money affected; Finance action `resolved`; amount 25.50. **Actions:** complete the clinician change, then the Finance adjustment; inspect the `request-refund` result and H2 rows; retry the same job only in a controlled test. **Expected/pass rule:** Finance decision precedes refund worker; returned status `SUCCESSFUL`, reference begins `RF-`, one matching payment-ledger row and one audit receipt; repeated

identical job does not add rows. Fail on refund before Finance authorisation or duplicate rows.

Actual/verdict: NOT EVIDENCED in a committed final-version engine run. Five A Java tests cover the isolated worker/database cases (E3).

Current-baseline automated verification

U01 Member A refund and investigation workers

Preconditions/data/actions: Java 21 and Maven; run `mvn clean test` from the formal module.

`RefundWorkersTest` uses an in-memory H2 database and mocked incoming jobs. It checks a resolved refund (25.50), pending refund (10.01), repeated resolved refund (12.00), investigation (10.00) and missing case ID.

Expected/pass rule: five tests pass with no failure/error; the tests assert ledger/audit status, reference and retry behaviour. **Actual:** five passed, zero failed/errors. **Verdict:** PASS at fc6293e. **Evidence:** E3. This does not prove a live provider refund.

U02 Member D care-change worker

Preconditions/data/actions: same Maven command; `CareChangeWorkersTest` uses an in-memory H2 database and mocked incoming jobs. It checks approved notification, unauthorised skip, Finance hand-off, invalid-input correction, retry receipt reuse, unknown target rejection and compatibility of B's original audit append. **Expected/pass rule:** seven tests pass and database assertions hold. **Actual:** seven passed, zero failed/errors. **Verdict:** PASS at fc6293e. **Evidence:** E3. This does not prove Tasklist form validation or a live notification.

U03 Member B booking worker and domain database

Preconditions/data/actions: use an isolated H2 database and mocked jobs or a labelled final-version process instance. Invoke `check-slot` for a first visit, `flag-resource-unavailable` for treatment, then `reserve-appointment` with the same case and time window; repeat the pending and ready jobs once. Query `booking_slot` by `occupancy_key` and `audit_event` by case. **Expected/pass rule:** one unique occupancy row per case/resource/window; unavailable treatment is PENDING with an incremented retry count; later ready status becomes CONFIRMED; repeats do not create a second occupancy row; worker variables agree with the stored state. **Actual/verdict:** NOT EVIDENCED. No B-specific automated test is committed. The current source's existing-row branch does not promote PENDING to CONFIRMED, so this is an identified code-level mismatch requiring a fix and rerun (E6). The older B engine observations in E5 use different model versions and do not close this check.

U04 Member C letter and referrer workers

Preconditions/data/actions: use an isolated H2 database and mocked jobs or labelled final-version process instances. Invoke `dispatch-clinic-letter` after P9 and `notify-referrer` after P2 redirect with non-empty case references; repeat each job once. Inspect returned references and `audit_event` by idempotency key.

Expected/pass rule: the letter returns `clinicLetterDispatchStatus=dispatched` and `C-DISPATCH-<case>`; redirect returns `referrerNotificationStatus=sent` and `C-REFERRER-<case>`; each action has exactly one audit row after retry, with no real external delivery claim. **Actual/verdict:** NOT EVIDENCED behaviourally. The mappings and keyed audit writes are present in E7, but no C-specific automated or final-version engine result is committed.

H01 Earlier member B route observations

Preconditions/data/actions: read E5's instance keys and recorded variable/end-state observations. The separate booking/payment model records no-pay, mock-success and mock-failure paths ending, plus an invalid-input incident and a repeated-payment instance left waiting at `PaymentReceived`. The earlier whole-hospital process version 1 records six routes, including completion and pending/urgent branches.

Expected/pass rule for current submission: use these records only as dated historical design evidence; do not label any of them a pass of the final BPMN. **Actual/verdict:** HISTORICAL OBSERVATIONS, not current acceptance. G07 was subsequently rerun and passed in E9; G09/G11 still need their own formal-baseline checks.

S01 BPMN structural alignment

Preconditions/data/actions: Python 3, formal BPMN and Java source available; run `python scripts/test_member_d_bpmn.py` from the formal module. **Expected/pass rule:** four checks pass; BPMN sequence-flow references and `zeebe:taskDefinition` types agree with the Java worker declarations. **Actual:** four tests passed. **Verdict:** PASS at fc6293e. **Evidence:** E4. Static checks do not prove deployment to a running cluster.

Open defects and required repeat checks

1. **Role authorisation is a confirmed failure.** G03 remains FAIL until PB-22 implements access control and the test is rerun with distinct role identities. A form field containing "Medical Secretary" is not sufficient.
2. **One complete final-model path is proven locally; wider coverage remains open.** E9 records the G07 REST run to COMPLETED, definition version, instance key, variables and worker logs. It has not yet received independent review or a final Git revision. Repeat G08-G11 for the other branches and add direct H2 row assertions and browser Tasklist form evidence where those claims matter.
3. **B's pending-slot database rule needs correction and a direct assertion.** G04 proves routing only. Source review shows that `check-slot` requests CONFIRMED even for a pending visit, while an existing PENDING row is not promoted to CONFIRMED by `reserve-appointment`. G09/U03 make both states measurable. No live database failure is claimed without a run.
4. **A successful mock payment is not a successful ledger write.** The older S1/S6 PDF expectations about a payment-ledger success row must not be presented as observed current behaviour. Check `message/variables` for payment and reserve ledger assertions for workers that write the ledger.
5. **Second-owner review remains open in the source PB-14/PB-18 documents.** The group should sign off the final version and retain the exact evidence paths in the repository before submission.
6. **B and C need owner-specific behavioural evidence.** Existing Java test output covers A and D only. Add B booking database tests and C mock-receipt/retry tests or labelled final-version engine runs, then update U03/U04 and attach logs. A's P8/P13 and D's P10 branches also need explicit final-version route records.